

Cryptographic Authentication for UAV Swarm Security: A Comparative Analysis of Detection Methods Against Spoofing Attacks

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Abstract—Unmanned Aerial Vehicles (UAVs) increasingly rely on Remote ID broadcast systems for airspace safety, yet these systems are vulnerable to spoofing attacks including phantom UAV injection, position falsification, and coordinated multi-UAV attacks. This paper presents a comprehensive evaluation of cryptographic versus graph-based detection methods for identifying malicious UAVs in swarm networks. We implement and evaluate four detection approaches: (1) Ed25519 digital signature verification, (2) spectral graph analysis, (3) centrality-based anomaly detection, and (4) machine learning with graph embeddings. Our experimental results demonstrate that cryptographic authentication achieves perfect detection (100% True Positive Rate, 0% False Positive Rate) for phantom and coordinated attacks across all mobility scenarios, while graph-based heuristic methods exhibit inconsistent performance with false positive rates ranging from 0% to 100%. For position falsification attacks, all methods show fundamental limitations rooted in the attack’s topology-preserving nature. We conclude that only cryptographic authentication provides provable security guarantees suitable for production UAV systems, with acceptable computational overhead (approximately 58ms per detection cycle for 30 UAVs). Our findings establish that cryptographic methods transform UAV security from a probabilistic problem (heuristics with variable detection rates) to a deterministic problem with mathematical security guarantees. The complete implementation is available as open-source software at <https://github.com/Sang-Buster/Argus>.

Index Terms—UAV security, Remote ID, cryptographic authentication, Ed25519, graph anomaly detection, spoofing attacks, swarm networks

I. INTRODUCTION

A. Motivation

The proliferation of Unmanned Aerial Vehicles (UAVs) in commercial and recreational airspace has necessitated the development of Remote ID broadcast protocols, which enable ground observers and other aircraft to identify and track nearby UAVs [1]. However, these broadcast-based identification systems are inherently vulnerable to spoofing attacks, where malicious actors inject false messages or manipulate transmitted data to deceive monitoring systems. Such attacks pose significant risks to airspace safety, privacy, and security [2], [3].

Remote ID systems face three primary attack vectors. First, phantom UAV injection occurs when attackers broadcast fabricated messages appearing to originate from non-existent UAVs, potentially overwhelming airspace manage-

ment systems or masking unauthorized operations. Second, position falsification involves compromised legitimate UAVs reporting false GPS coordinates, enabling unauthorized flight in restricted zones while appearing compliant. Third, coordinated multi-UAV attacks employ multiple compromised UAVs executing synchronized spoofing to achieve objectives beyond single-UAV capabilities. These threats collectively undermine the fundamental trust assumptions of broadcast-based identification systems.

B. Related Work

Prior research on UAV security has explored various detection approaches:

Graph-based methods leverage network topology analysis for anomaly detection. Spectral graph theory [4], [5] has been applied to detect structural anomalies through eigenvalue analysis of the graph Laplacian. Centrality-based approaches [6] identify suspicious nodes based on abnormal betweenness, degree, or closeness centrality metrics.

Machine learning approaches employ supervised and unsupervised learning on graph features. Node2Vec [7] generates graph embeddings through random walks, enabling anomaly detection via classification or clustering algorithms.

Cryptographic methods provide authentication through digital signatures. Ed25519 [8], [9], based on elliptic curve cryptography, offers fast signing and verification suitable for resource-constrained embedded systems. Hardware acceleration techniques [10] further improve performance for embedded deployments.

However, **no comprehensive comparison** exists evaluating cryptographic versus graph-based approaches specifically for UAV Remote ID security across multiple attack scenarios and mobility conditions.

C. Contributions

This work presents the first systematic evaluation of cryptographic authentication in conjunction with graph-based detection methods for UAV Remote ID spoofing attacks. We implement and compare asymmetric cryptographic authentication using Ed25519 with spectral graph analysis, centrality-based heuristics, and machine learning based anomaly detection across multiple attack scenarios and mobility conditions.

The computational overhead associated with cryptographic verification is quantified, demonstrating deterministic detection despite higher cost. Finally, we release an open-source framework, Argus, enabling reproducible experimentation and standardized evaluation of UAV security mechanisms.

D. Paper Organization

Section II describes our threat model and system architecture. Section III details the four detection methods. Section IV presents experimental methodology. Section V reports comprehensive results across three attack types and two mobility scenarios. Section VI analyzes the findings and discusses implications. Section X concludes with recommendations for production deployment.

II. SYSTEM MODEL AND THREAT MODEL

A. UAV Swarm Network Model

We model a UAV swarm [11] as a dynamic communication graph $G(t) = (V(t), E(t))$ where vertices $V(t) = \{v_1, v_2, \dots, v_n\}$ represent n UAVs at time t , and edges $E(t) \subseteq V(t) \times V(t)$ represent bidirectional communication links. Edge existence is determined by Euclidean distance such that $(v_i, v_j) \in E(t)$ if and only if $\|p_i(t) - p_j(t)\| \leq r_{comm}$, where $\|\cdot\|$ denotes the Euclidean norm, $p_i(t) \in \mathbb{R}^3$ denotes the position of UAV i at time t , and r_{comm} represents the communication range.

Remote ID Messages: Each UAV broadcasts periodic messages containing:

```
RemoteID_Message = {
  uav_id: String,
  position: (x, y, z) in R^3,
  velocity: (vx, vy, vz) in R^3,
  timestamp: Unix_Time,
  signature: Bytes[64] (optional)
}
```

B. Threat Model

We consider three classes of spoofing attacks against UAV Remote ID systems.

Phantom UAV Injection (A_1) occurs when an adversary injects k fabricated Remote ID messages that appear to originate from non-existent UAVs. Formally, at time t_{attack} , the swarm vertex set is modified as

$$V(t_{attack}) \leftarrow V(t_{attack}) \cup \{phantom_1, \dots, phantom_k\} \quad (1)$$

where phantom UAVs lack valid cryptographic credentials.

Position Falsification (A_2) involves compromising legitimate UAVs, causing them to report false positions while maintaining valid cryptographic signatures. For a compromised UAV i , the reported position is given by

$$\forall i \in Compromised : p'_i(t) = p_i(t) + \delta_i \quad (2)$$

where δ_i is a position offset vector and $\|\delta_i\| \leq M_{max}$ (maximum falsification magnitude).

Coordinated Attacks (A_3) extend phantom injection by introducing multiple malicious transmitters arranged in a coordinated geometric pattern (e.g., circular formation) to achieve specific spatial coverage or deception objectives.

The adversary is assumed to be capable of broadcasting arbitrary RF signals in the Remote ID frequency band, capturing and replaying legitimate messages, compromising UAV flight controllers to enable falsified position reporting, and coordinating multiple attack nodes. However, we assume that the adversary cannot extract private keys from secure hardware elements, cannot break the cryptographic assumptions underlying Ed25519, and remains subject to physical constraints such as RF propagation and timing requirements.

C. System Architecture

Our experimental framework (Argus) consists of five main components: UAV Swarm Simulation (physics, communications, mobility), Attack Injection Module (phantom, position, coordinated), Communication Graph construction $G(t) = (V, E)$, Crypto Detection (Ed25519, PKI) and Graph-Based Detection (spectral, centrality, ML), and Evaluation & Metrics (TPR, FPR, F1, detection time, reproducibility).

III. DETECTION METHODS

We implement and evaluate four detection approaches representing the state-of-the-art in UAV security.

A. Cryptographic Authentication

Principle: Verify message authenticity through Ed25519 digital signatures based on elliptic curve cryptography over Curve25519.

Algorithm 1 Cryptographic Detection

Require: Graph $G(t)$, Public Key Database PKI

Ensure: Set of anomalous UAV IDs

Initialize: $Anomalous \leftarrow \emptyset$

for each node $v \in V(G)$ **do**

$msg \leftarrow \text{latest_message}(v)$

if $msg.signature = \text{NULL}$ **then**

$Anomalous \leftarrow Anomalous \cup \{v\}$ {No signature}

else if $v \notin \text{PKI}$ **then**

$Anomalous \leftarrow Anomalous \cup \{v\}$ {Unknown UAV}

else if $\neg \text{Verify_Ed25519}(msg, PKI[v])$ **then**

$Anomalous \leftarrow Anomalous \cup \{v\}$ {Invalid signature}

end if

end for

return $Anomalous$

Key Generation: Each legitimate UAV generates an Ed25519 keypair (sk_i, pk_i) where the private key $sk_i \in \{0, 1\}^{256}$ is stored securely on-board and the public key pk_i is registered in the system's PKI during initialization.

Message Signing: UAV i signs each Remote ID message m (containing position, velocity, timestamp) as:

$$\sigma_i = \text{Ed25519.Sign}(sk_i, m) \quad (3)$$

ID messages at 1 Hz. Each simulation runs for 50 timesteps to capture attack injection, detection, and recovery phases.

Mobility Model: We consider two mobility scenarios to assess detector robustness. In stationary mode, UAVs remain at fixed positions throughout the simulation. In mobile mode, UAVs move toward randomly selected destinations at velocities of $v = 10 \pm 2$ m/s, with new destinations assigned upon arrival to simulate realistic patrol or surveillance patterns.

Reproducibility: All experiments use fixed random seed (seed=42) and standardized configuration parameters to ensure reproducibility.

B. Attack Scenarios

Scenario 1: Phantom UAV Attack involves injecting $k = 3$ phantom UAVs at $t = 10$ seconds for a duration of 20 seconds. Phantom positions are randomly selected within communication range of legitimate UAVs to maximize network integration and detection difficulty.

Scenario 2: Position Falsification simulates the compromise of $m = 4$ legitimate UAVs (approximately 13% of the swarm), each reporting false positions offset by $\|\delta\| = 100$ meters in random directions. This magnitude represents significant spatial deception while maintaining communication topology.

Scenario 3: Coordinated Attack tests organized threat patterns by injecting $k = 5$ phantom UAVs arranged in a circular formation with radius 50 meters, simulating sophisticated adversaries capable of geometric coordination.

C. Evaluation Metrics

For each detector D and attack scenario A , we compute:

True Positive Rate (Recall):

$$TPR = \frac{TP}{TP + FN} \quad (7)$$

False Positive Rate:

$$FPR = \frac{FP}{FP + TN} \quad (8)$$

Precision:

$$Precision = \frac{TP}{TP + FP} \quad (9)$$

F1 Score (harmonic mean):

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (10)$$

Detection Time: Average time per detection cycle (milliseconds)

where TP = true positives (malicious UAVs correctly identified), FP = false positives (legitimate UAVs incorrectly flagged), TN = true negatives, FN = false negatives.

D. Experimental Procedure

Each experiment follows a standardized protocol: initialize swarm with n UAVs, collect 30 timesteps of baseline graphs, train detectors, inject attack at $t = 10$ s, run detection at every timestep, remove attack at $t = 30$ s, and continue for 20 timesteps to observe recovery. Each configuration uses controlled random seeds (seed=42) for reproducibility.

V. EXPERIMENTAL RESULTS

A. Stationary Swarm Results

Table I presents detection performance for stationary swarms (30 UAVs, $r_{comm} = 200$ m).

Key Observations: The results validate our primary hypothesis that cryptographic detection achieves perfect performance (TPR=1.0, FPR=0.0, F1=1.0) for phantom and coordinated attacks. Notably, spectral detection also achieves perfect performance for these attack types, demonstrating that topology-based methods can be effective when attacks create structural anomalies. Position falsification proves undetectable by topology-preserving methods (spectral, crypto) as expected, since falsifying position without changing communication links creates no graph structural changes. Centrality and ML detectors exhibit unacceptable false positive rates (87–100%), rendering them impractical for production deployment. Regarding computational overhead, cryptographic verification requires approximately $30\times$ more time than spectral analysis (58ms vs 2ms) but remains acceptable for real-time operation at 1 Hz broadcast rates.

B. Mobility Impact Analysis

Table II compares detector performance in mobile swarms where UAVs move at 10 m/s.

Key Observations: Cryptographic detection exhibits complete mobility invariance, maintaining perfect performance (100% TPR, 0% FPR) regardless of UAV movement patterns. This confirms that signature verification operates independently of network dynamics. In contrast, spectral detection degrades significantly under mobility, with false positive rates increasing from 0% to 62.5% for phantom attacks. This degradation occurs because movement-induced topology changes create false anomaly signals that spectral methods cannot distinguish from genuine attacks. The sensitivity of graph-based methods to dynamics manifests as shifts in eigenvalue distributions, necessitating mobility-aware threshold adaptation for practical deployment.

C. Detection Time Analysis

Table III presents the computational overhead comparison across all detection methods. Cryptographic verification incurs approximately $30\times$ computational overhead compared to spectral analysis, primarily due to Ed25519 signature verification for all nodes, PKI lookups, and message parsing overhead.

For a 30-UAV swarm at 1 Hz broadcast rate, total overhead is 58ms per detection cycle, well within real-time constraints (< 1000 ms budget).

Trade-off: Pay 56ms additional latency for 100% security guarantee (TPR=1.0, FPR=0.0) versus probabilistic heuristics.

VI. ANALYSIS AND DISCUSSION

A. Why Cryptographic Methods Succeed

Ed25519 security [8], [9] relies on the discrete logarithm problem: computing private key sk from public key $pk = [sk]G$ requires $O(2^{128})$ operations. Phantom UAVs cannot

TABLE I
DETECTION PERFORMANCE IN STATIONARY SWARMS

Attack Type	Detector	TPR	FPR	Precision	Recall	F1	Time (ms)
Phantom	Spectral	1.000	0.000	1.000	1.000	1.000	1.97
	Centrality	0.667	1.000	0.062	0.667	0.114	1.24
	Crypto	1.000	0.000	1.000	1.000	1.000	57.90
	ML	0.333	0.933	0.034	0.333	0.062	4.88
Position	Spectral	0.000	0.000	0.000	0.000	0.000	1.75
	Centrality	1.000	1.000	0.100	1.000	0.182	1.12
	Crypto	0.000	0.000	0.000	0.000	0.000	57.27
	ML	0.667	0.963	0.071	0.667	0.129	4.63
Coordinated	Spectral	1.000	0.000	1.000	1.000	1.000	1.90
	Centrality	1.000	1.000	0.091	1.000	0.167	1.51
	Crypto	1.000	0.000	1.000	1.000	1.000	57.21
	ML	1.000	0.867	0.103	1.000	0.188	5.11

TABLE II
STATIONARY VS MOBILE SWARM PERFORMANCE

Attack	Detector	Stationary (TPR/FPR)	Mobile (TPR/FPR)
Phantom	Spectral	100.0%/0.0%	99.2%/62.5%
	Crypto	100.0%/0.0%	100.0%/0.0%
Position	Spectral	0.0%/0.0%	1.9%/3.2%
	Crypto	0.0%/0.0%	0.0%/0.0%
Coordinated	Spectral	100.0%/0.0%	99.0%/6.7%
	Crypto	100.0%/0.0%	100.0%/0.0%

TABLE III
COMPUTATIONAL OVERHEAD COMPARISON

Detector	Mean (ms)	Std Dev (ms)	Overhead
Spectral	1.87	0.12	1.0×
Centrality	1.29	0.17	0.7×
Crypto	57.46	2.31	30.7×
ML	4.87	0.51	2.6×

forge valid signatures without the private key. While graph-based methods detect anomalies probabilistically (requiring threshold tuning), cryptographic methods provide **deterministic guarantees**: valid signatures prove authenticity with probability $1 - \epsilon$ where $\epsilon < 2^{-128}$.

B. Why Graph-Based Methods Fail or Degrade

Configuration Sensitivity: Spectral and centrality detectors require threshold calibration specific to spatial domain, communication range, and swarm size. Our experiments show $500 \times 500\text{m}$ requires spectral threshold 4.0 while $1000 \times 1000\text{m}$ requires 2.5.

Mobility Vulnerability: Movement creates dynamic topology changes that shift baseline eigenvalue distributions, create transient centrality anomalies, and generate false positives when benign changes are mistaken for attacks.

Evasion Potential: Adaptive attackers [15] can craft attacks that blend with normal variations by injecting phantoms in low-density regions or operating below detection thresholds.

Graph methods fundamentally detect topology anomalies; attacks preserving topology are inherently undetectable.

C. Position Falsification: A Special Case

All detectors achieve 0% TPR on position falsification (Table I). When compromised UAVs report $p'_i = p_i + \delta$ with $\|\delta\| < r_{comm}$, the communication graph remains unchanged: $G(p') = G(p) \implies L(p') = L(p)$. Position falsification is **topologically invisible**. Cryptographic signatures prove message authenticity, not data truthfulness—compromised UAVs can sign false positions with legitimate keys.

Mitigation (future work): Multi-lateration [16], [17] could verify positions against distance measurements; physics-based validation could detect impossible dynamics; Byzantine consensus [18], [19] could require $\lfloor n/3 \rfloor + 1$ agreement.

D. ML Detector Performance Analysis

The ML detector exhibits high FPR (87–96%) despite perfect recall for phantom/coordinated attacks. Root causes include: contamination mismatch (configured for 15% anomalies vs. actual $\sim 9\%$ attack density), limited 4D feature vector compared to spectral eigenvectors, and potential overfitting on 30 baseline graphs. Tuning contamination did not substantially improve results, suggesting fundamental limitations. Future work should explore Graph Neural Networks and temporal models for improved feature learning.

VII. SUMMARY OF CONTRIBUTIONS

A. Novel Contributions

This work makes three novel contributions to UAV security research:

1. First Comprehensive Cryptographic Comparison: Prior work focuses exclusively on either cryptographic or graph-based approaches. We provide the first systematic evaluation demonstrating $30\times$ computational overhead, mobility impact assessment (crypto invariant vs. heuristics degrading), and reproducibility analysis.

2. Threshold Scaling Analysis: We identify and characterize the spatial scaling problem where optimal thresholds depend on spatial domain size, communication range, and swarm

size. Small domains (500×500m) require spectral threshold 4.0, while large domains (1000×1000m) require threshold 2.5.

3. Mobility Impact Quantification: We quantify mobility-induced degradation, showing spectral FPR increases from 0% to 62.5% for phantom attacks while cryptographic FPR remains 0%.

B. Practical Implications

The Argus framework provides unified attack simulation, multi-modal mobility, standardized evaluation with reproducible seeds, and real-time visualization. For practitioners, our results establish Ed25519 as suitable for real-time Remote ID (58ms latency) and demonstrate heuristic-only approaches are insufficient for security-critical applications.

VIII. DISCUSSION

A. Cryptographic vs Heuristic Security

Our results establish a fundamental distinction. **Heuristic methods** provide probabilistic detection with performance depending on configuration, spatial scale, and mobility. Adaptive attackers can potentially evade detection by crafting attacks below thresholds. **Cryptographic methods** provide deterministic guarantees with detection probability $1 - 2^{-128}$, invariant to scale and mobility. Evasion requires solving the discrete logarithm problem—computationally infeasible. The 30× overhead (58ms) purchases provable security.

Recommendation: For security-critical UAV applications, cryptographic authentication is **mandatory**. Heuristics may supplement but cannot replace cryptographic mechanisms.

B. Hybrid Approach and Limitations

A hybrid detector (crypto filtering followed by spectral analysis) does not improve position attack detection since falsification with $\|\delta\| < r_{comm}$ creates no topology changes. Alternative approaches (multi-lateration [17], physics validation, Byzantine consensus [18]) are required.

Limitations: All methods fail on topology-preserving position attacks. Cryptographic methods cannot detect insider threats with valid keys. The 30× overhead may limit scalability beyond 1000 UAVs without hardware acceleration. Deployment requires PKI infrastructure.

C. Threat Landscape

Current threats (phantom injection, coordinated attacks) are addressed with 100% detection. Emerging threats remain: GPS spoofing [20], [21] affects all UAVs simultaneously; RF jamming creates denial-of-service; AI-powered adaptive attacks [15] could learn thresholds; quantum computing threatens Ed25519 via Shor’s algorithm [22]. Post-quantum alternatives (CRYSTALS-Dilithium [23], Falcon [24]) should be considered for long-term deployments.

IX. FUTURE WORK

A. Short-Term Extensions

Near-term extensions include implementing position verification via multi-lateration [16], [17] using signal strength and time-of-arrival measurements, developing mobility-aware threshold adaptation functions for heuristic detectors, and replacing Isolation Forest with Graph Neural Networks [25], [26] such as GCN or GAT for improved feature learning.

B. Long-Term Directions

Long-term research should evaluate quantum-resistant signatures (CRYSTALS-Dilithium [23], Falcon [24]), implement Byzantine Fault Tolerance protocols [18], [19] for distributed position verification, and develop hardware-accelerated [10] Ed25519 using GPU or FPGA for sub-millisecond latency in large-scale swarms.

X. CONCLUSION

This paper presents the first comprehensive comparison of cryptographic and graph-based detection methods for UAV spoofing attacks. Through extensive experimentation across three attack types (phantom injection, position falsification, coordinated attacks) and two mobility scenarios, we establish that Ed25519 cryptographic authentication achieves **perfect detection** (TPR=1.0, FPR=0.0, F1=1.0) for phantom and coordinated attacks, while graph-based heuristic methods exhibit inconsistent performance with false positive rates up to 100%.

Our evaluation reveals four critical findings: (1) only cryptographic methods provide security guarantees based on computational hardness assumptions; (2) mobility invariance distinguishes cryptographic approaches (zero degradation) from graph-based methods (FPR increases up to 62.5%); (3) position falsification remains an open problem requiring multi-modal sensor fusion; (4) computational overhead proves acceptable with 58ms detection latency meeting real-time requirements.

We recommend deploying Ed25519 cryptographic authentication as the **mandatory** primary security mechanism for all production UAV systems. Spectral analysis may supplement as a monitoring tool but should not serve security-critical functions. Our open-source Argus framework enables reproducible research at <https://github.com/Sang-Buster/Argus>.

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APPENDIX

All experiments use Python 3.10 with NetworkX, NumPy, scikit-learn, and pycryptodome, with fixed random seed

(seed=42). Complete source code and experiment scripts are available in the Argus repository.

Reproduction Instructions:

```
git clone https://github.com/Sang-Buster/Argus
cd Argus
uv sync # Install dependencies
uv run argus --attack phantom \
  --detectors all --mode comparison \
  --num-uavs 30
```

The deterministic output is shown in Table IV, which demonstrates cryptographic detection’s linear scalability $O(n)$, with detection time increasing proportionally from 39.12ms (20 UAVs) to 57.90ms (30 UAVs) while maintaining perfect accuracy.

TABLE IV
SWARM SIZE IMPACT (PHANTOM ATTACK, CRYPTO DETECTOR)

UAVs (n)	TPR	FPR	F1	Time (ms)
20	1.000	0.000	1.000	39.12
25	1.000	0.000	1.000	47.87
30	1.000	0.000	1.000	57.90

Table V illustrates spectral detection’s extreme threshold sensitivity. While $\tau^* = 2.5$ achieves perfect detection, minor deviations cause catastrophic degradation: $\tau = 2.0$ introduces 26.7% false positives, $\tau = 3.0$ reduces TPR to 66.7%, and $\tau = 4.0$ eliminates all detections. This brittleness necessitates environment-specific calibration and enables threshold-evading attacks.

TABLE V
SPECTRAL THRESHOLD SENSITIVITY (PHANTOM ATTACK, 30 UAVs)

Threshold (τ)	TPR	FPR	F1
2.0	1.000	0.267	0.600
2.5	1.000	0.000	1.000
3.0	0.667	0.000	0.800
4.0	0.000	0.000	0.000

The Argus framework comprises approximately 3,500 lines of Python code organized into modular subsystems: core simulation (~ 800 LOC), cryptographic operations (~ 150 LOC), detection algorithms ($\sim 1,200$ LOC), attack injection (~ 600 LOC), and evaluation/visualization (~ 400 LOC). The framework achieves 85% test coverage with emphasis on cryptographic operations and attack injection.

Memory consumption for 30 UAVs totals approximately 15 MB (swarm state), 2 MB per timestep (graphs), and 60 MB (baseline samples), with 200 MB peak including visualization. Execution time for a complete comparison run (30 UAVs, 50 timesteps) is approximately 15 seconds. Algorithmic complexity follows theoretical bounds: $O(n)$ for cryptographic verification, $O(n^2)$ for spectral eigendecomposition, and $O(n^3)$ for betweenness centrality.